

**(C1)** *Identify the decision alternatives, the states of nature, the uncertain event, and the payoff components.* Then **(C2)** *construct the corresponding payoff table.*

## Context Descriptions

1. A school event company must choose between a compact snack stand and a full snack stand before attendance is known. Attendance may be high or low. Analysts have already converted each possible consequence into direct profit measured in thousand USD. The compact stand produces a profit of 18 thousand USD under high attendance and 11 thousand USD under low attendance. The full stand produces a profit of 27 thousand USD under high attendance and 6 thousand USD under low attendance.
2. A weekend pop-up business must choose between an indoor booth and a mobile stall before market demand is known. Demand may be strong or weak. The indoor booth contributes a fixed 550 thousand JPY to profit, while the mobile stall contributes a fixed 420 thousand JPY. Strong demand adds a fixed market bonus of 180 thousand JPY to either alternative. Weak demand instead reduces profit by a fixed 90 thousand JPY because of permit and cleanup expenses.
3. A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic. Clear traffic also adds a fixed scheduling bonus of 5 thousand CNY to either fleet, while congestion creates a fixed loss of 4 thousand CNY.
4. A city-tour operator must choose between a standard tour and a premium tour before tourism demand is known. Demand may be high or low. The standard tour contributes 2 thousand EUR per group booking and also provides a fixed contract contribution of 6 thousand EUR. The premium tour contributes 4 thousand EUR per group booking and provides a fixed contract contribution of 10 thousand EUR. High tourism demand adds another 3 thousand EUR per group booking to either tour, while low demand adds 1 thousand EUR per group booking. The standard tour is expected to complete 16 group bookings under high demand and 10 under low demand; the premium tour is expected to complete 12 group bookings under high demand and 7 under low demand.
5. A solar-installation company must choose between a standard installation crew and an expanded installation crew before customer orders are known. Orders may be strong, regular, or weak. The standard crew contributes 3 thousand GBP per completed installation and provides a fixed contract contribution of 8 thousand GBP. The expanded crew contributes 4 thousand GBP per completed installation and provides a fixed contract contribution of 5 thousand GBP. Strong orders add 2 thousand GBP per completed installation and a fixed scheduling bonus of 6 thousand GBP. Regular orders add 1 thousand GBP per completed installation and no fixed adjustment. Weak orders reduce the contribution by 1 thousand GBP per completed installation and create a fixed idle-time loss of 4 thousand GBP. The standard crew is expected to complete 10 installations under strong orders, 7 under regular orders, and 4 under weak orders. The expanded crew is expected to complete 14, 9, and 5 installations under the same respective states.
6. A coffee exporter must choose among contracted storage, a rented warehouse, and an automated distribution hub before export demand and shipping conditions are known. Conditions may be favorable, normal, or adverse. Contracted storage contributes 4 million COP per export lot handled but requires a fixed storage commitment of 12 million COP. The rented warehouse contributes 5 million COP per export lot but requires a fixed commitment of 20 million COP. The automated hub contributes 6 million COP per export lot but requires a fixed commitment of 32 million COP. Favorable conditions add 2 million COP per export lot and a fixed logistics bonus of 8 million COP. Normal conditions make no additional per-lot or fixed adjustment. Adverse conditions reduce the contribution by 2 million COP per export lot and create a fixed disruption loss of 10 million COP. Contracted storage is expected to handle 18 export lots under favorable conditions, 13 under normal conditions, and 8 under adverse conditions. The rented warehouse is expected to handle 22, 15, and 9 export lots, while the automated hub is expected to handle 26, 17, and 10 export lots under the same respective states.

1 Solutions: Extraction

| Problems:                                   |
|---------------------------------------------|
| 1.1. School Event Snack Stand (C1–C2)       |
| 1.2. Weekend Pop-Up Business (C1–C2)        |
| 1.3. Local Delivery Fleet (C1–C2)           |
| 1.4. City-Tour Format (C1–C2)               |
| 1.5. Solar-Installation Crew (C1–C2)        |
| 1.6. Coffee-Export Storage Strategy (C1–C2) |

1.1 Exercise 1: School event snack stand

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                           |
|------------------|-------------------------------------------------------|
| Alternatives     | <b>A1:</b> Compact stand <b>A2:</b> Full stand        |
| States of nature | <b>S1:</b> High attendance <b>S2:</b> Low attendance  |
| Events           | Attendance level after the stand is selected          |
| Consequences     | Profit in thousand USD for each stand–attendance pair |



C2 - Building the payoff table. Relevant values. The context supplies a direct profit for every stand–attendance pair.

| Abstract component | Context component                                        | Math symbol        | Corresponding value                                                                                               |
|--------------------|----------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------|
| Direct payoff      | Profit stated for each stand under each attendance level | $P_{ay}(A_i, S_j)$ | Compact–high: 18 thousand USD; Compact–low: 11 thousand USD; Full–high: 27 thousand USD; Full–low: 6 thousand USD |

**Payoff equation.** Because these are direct-payoff data,  $P_{ay}(A_i, S_j)$  is read directly from the context.

Calculations.

| Stand         | High attendance          | Low attendance           |
|---------------|--------------------------|--------------------------|
| Compact stand | Direct context value: 18 | Direct context value: 11 |
| Full stand    | Direct context value: 27 | Direct context value: 6  |

Final payoff table (profit in thousand USD).

| Stand         | High attendance | Low attendance |
|---------------|-----------------|----------------|
| Compact stand | 18              | 11             |
| Full stand    | 27              | 6              |

1.2 Exercise 2: Weekend pop-up business

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                        |                         |
|------------------|----------------------------------------------------|-------------------------|
| Alternatives     | <b>A1:</b> Indoor booth                            | <b>A2:</b> Mobile stall |
| States of nature | <b>S1:</b> Strong demand                           | <b>S2:</b> Weak demand  |
| Events           | Market demand after the selling format is selected |                         |
| Consequences     | Profit in thousand JPY for each format–demand pair |                         |



C2 - Building the payoff table. Relevant values.

| Abstract component           | Context component                                                             | Math symbol | Corresponding value                                            |
|------------------------------|-------------------------------------------------------------------------------|-------------|----------------------------------------------------------------|
| Fixed, alternative dependent | Fixed contribution to profit from the chosen selling format                   | $f^A(A_i)$  | Indoor booth: 550 thousand JPY; Mobile stall: 420 thousand JPY |
| Fixed, state dependent       | Market bonus under strong demand or permit-and-cleanup loss under weak demand | $f^S(S_j)$  | Strong demand: 180 thousand JPY; Weak demand: −90 thousand JPY |

Payoff equation.

$$P_{ay}(A_i, S_j) = f^A(A_i) + f^S(S_j).$$

Calculations (thousand JPY).

| Format       | Strong demand     | Weak demand         |
|--------------|-------------------|---------------------|
| Indoor booth | $550 + 180 = 730$ | $550 + (-90) = 460$ |
| Mobile stall | $420 + 180 = 600$ | $420 + (-90) = 330$ |

Final payoff table (profit in thousand JPY).

| Format       | Strong demand | Weak demand |
|--------------|---------------|-------------|
| Indoor booth | 730           | 460         |
| Mobile stall | 600           | 330         |

1.3 Exercise 3: Local delivery fleet

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                        |                               |
|------------------|----------------------------------------------------|-------------------------------|
| Alternatives     | <b>A1:</b> Bicycle fleet                           | <b>A2:</b> Electric-van fleet |
| States of nature | <b>S1:</b> Clear traffic                           | <b>S2:</b> Congested traffic  |
| Events           | Traffic conditions after the fleet is selected     |                               |
| Consequences     | Profit in thousand CNY for each fleet–traffic pair |                               |



C2 - Building the payoff table. Relevant values.

| Abstract component                    | Context component                                                               | Math symbol   | Corresponding value                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| Quantity, alternative–state dependent | Expected completed delivery batches for each fleet under each traffic condition | $q(A_i, S_j)$ | Bicycle–clear: 14 batches; Bicycle–congested: 10 batches; Electric van–clear: 12 batches; Electric van–congested: 8 batches |
| Variable, alternative dependent       | Earnings per completed delivery batch for the chosen fleet                      | $v^A(A_i)$    | Bicycle fleet: 4 thousand CNY/batch; Electric-van fleet: 6 thousand CNY/batch                                               |
| Fixed, state dependent                | Scheduling bonus in clear traffic or congestion loss                            | $f^S(S_j)$    | Clear traffic: 5 thousand CNY; Congested traffic: −4 thousand CNY                                                           |

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Calculations (thousand CNY).

| Fleet              | Clear traffic    | Congested traffic   |
|--------------------|------------------|---------------------|
| Bicycle fleet      | $14(4) + 5 = 61$ | $10(4) + (-4) = 36$ |
| Electric-van fleet | $12(6) + 5 = 77$ | $8(6) + (-4) = 44$  |

Final payoff table (profit in thousand CNY).

| Fleet              | Clear traffic | Congested traffic |
|--------------------|---------------|-------------------|
| Bicycle fleet      | 61            | 36                |
| Electric-van fleet | 77            | 44                |

1.4 Exercise 4: City-tour format

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                      |                               |
|------------------|--------------------------------------------------|-------------------------------|
| Alternatives     | <b>A1:</b> Standard tour                         | <b>A2:</b> Premium tour       |
| States of nature | <b>S1:</b> High tourism demand                   | <b>S2:</b> Low tourism demand |
| Events           | Tourism demand after the tour format is selected |                               |
| Consequences     | Profit in thousand EUR for each tour–demand pair |                               |



C2 - Building the payoff table. Relevant values.

| Abstract component                    | Context component                                                     | Math symbol   | Corresponding value                                                                                       |
|---------------------------------------|-----------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------|
| Quantity, alternative–state dependent | Expected group bookings for each tour under each tourism-demand level | $q(A_i, S_j)$ | Standard–high: 16 bookings; Standard–low: 10 bookings; Premium–high: 12 bookings; Premium–low: 7 bookings |
| Variable, alternative dependent       | Contribution per group booking from the chosen tour                   | $v^A(A_i)$    | Standard tour: 2 thousand EUR/booking; Premium tour: 4 thousand EUR/booking                               |
| Variable, state dependent             | Additional contribution per group booking from tourism demand         | $v^S(S_j)$    | High demand: 3 thousand EUR/booking; Low demand: 1 thousand EUR/booking                                   |
| Fixed, alternative dependent          | Fixed contract contribution from the chosen tour                      | $f^A(A_i)$    | Standard tour: 6 thousand EUR; Premium tour: 10 thousand EUR                                              |

Payoff equation.

$$Pay(A_i, S_j) = q(A_i, S_j) [v^A(A_i) + v^S(S_j)] + f^A(A_i).$$

Calculations (thousand EUR).

| Tour          | High tourism demand   | Low tourism demand   |
|---------------|-----------------------|----------------------|
| Standard tour | $16(2 + 3) + 6 = 86$  | $10(2 + 1) + 6 = 36$ |
| Premium tour  | $12(4 + 3) + 10 = 94$ | $7(4 + 1) + 10 = 45$ |

Final payoff table (profit in thousand EUR).

| Tour          | High tourism demand | Low tourism demand |
|---------------|---------------------|--------------------|
| Standard tour | 86                  | 36                 |
| Premium tour  | 94                  | 45                 |

1.5 Exercise 5: Solar-installation crew

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                          |                                       |                        |
|------------------|------------------------------------------------------|---------------------------------------|------------------------|
| Alternatives     | <b>A1:</b> Standard installation crew                | <b>A2:</b> Expanded installation crew |                        |
| States of nature | <b>S1:</b> Strong orders                             | <b>S2:</b> Regular orders             | <b>S3:</b> Weak orders |
| Events           | Customer-order level after the crew size is selected |                                       |                        |
| Consequences     | Profit in thousand GBP for each crew–order pair      |                                       |                        |



C2 - Building the payoff table. Relevant values.

| Abstract component                    | Context component                                                     | Math symbol   | Corresponding value                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantity, alternative–state dependent | Expected completed installations for each crew under each order level | $q(A_i, S_j)$ | Standard–strong: 10;<br>Standard–regular: 7;<br>Standard–weak: 4 installations;<br>Expanded–strong: 14;<br>Expanded–regular: 9;<br>Expanded–weak: 5 installations |
| Variable, alternative dependent       | Contribution per completed installation from the chosen crew          | $v^A(A_i)$    | Standard crew: 3 thousand GBP/installation; Expanded crew: 4 thousand GBP/installation                                                                            |
| Variable, state dependent             | Per-installation addition or reduction from the order level           | $v^S(S_j)$    | Strong: 2 thousand GBP/installation; Regular: 1 thousand GBP/installation; Weak: –1 thousand GBP/installation                                                     |
| Fixed, alternative dependent          | Fixed contract contribution from the chosen crew                      | $f^A(A_i)$    | Standard crew: 8 thousand GBP; Expanded crew: 5 thousand GBP                                                                                                      |
| Fixed, state dependent                | Scheduling bonus, no regular-order adjustment, or idle-time loss      | $f^S(S_j)$    | Strong: 6 thousand GBP; Regular: 0 thousand GBP; Weak: –4 thousand GBP                                                                                            |

Payoff equation.

$$Pay(A_i, S_j) = q(A_i, S_j) [v^A(A_i) + v^S(S_j)] + f^A(A_i) + f^S(S_j).$$

Calculations (thousand GBP).

| Crew                       | Strong orders            | Regular orders          | Weak orders                   |
|----------------------------|--------------------------|-------------------------|-------------------------------|
| Standard installation crew | $10(3 + 2) + 8 + 6 = 64$ | $7(3 + 1) + 8 + 0 = 36$ | $4(3 + (-1)) + 8 + (-4) = 12$ |
| Expanded installation crew | $14(4 + 2) + 5 + 6 = 95$ | $9(4 + 1) + 5 + 0 = 50$ | $5(4 + (-1)) + 5 + (-4) = 16$ |

Final payoff table (profit in thousand GBP).

| Crew                       | Strong orders | Regular orders | Weak orders |
|----------------------------|---------------|----------------|-------------|
| Standard installation crew | 64            | 36             | 12          |
| Expanded installation crew | 95            | 50             | 16          |

1.6 Exercise 6: Coffee-export storage strategy

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                                                  |                              |                                       |
|------------------|------------------------------------------------------------------------------|------------------------------|---------------------------------------|
| Alternatives     | <b>A1:</b> Contracted storage                                                | <b>A2:</b> Rented warehouse  | <b>A3:</b> Automated distribution hub |
| States of nature | <b>S1:</b> Favorable conditions                                              | <b>S2:</b> Normal conditions | <b>S3:</b> Adverse conditions         |
| Events           | Export-demand and shipping conditions after the storage strategy is selected |                              |                                       |
| Consequences     | Profit in million COP for each storage-condition pair                        |                              |                                       |



C2 - Building the payoff table. Relevant values.

| Abstract component                    | Context component                                                          | Math symbol   | Corresponding value                                                                                                                                             |
|---------------------------------------|----------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantity, alternative-state dependent | Expected export lots handled by each storage strategy under each condition | $q(A_i, S_j)$ | Contracted-favorable: 18; normal: 13; adverse: 8 lots; Rented-favorable: 22; normal: 15; adverse: 9 lots; Automated-favorable: 26; normal: 17; adverse: 10 lots |
| Variable, alternative dependent       | Contribution per export lot from the chosen storage strategy               | $v^A(A_i)$    | Contracted: 4 million COP/lot; Rented: 5 million COP/lot; Automated: 6 million COP/lot                                                                          |
| Variable, state dependent             | Per-lot addition, no adjustment, or reduction from conditions              | $v^S(S_j)$    | Favorable: 2 million COP/lot; Normal: 0 million COP/lot; Adverse: -2 million COP/lot                                                                            |
| Fixed, alternative dependent          | Fixed storage commitment required by the chosen strategy                   | $f^A(A_i)$    | Contracted: -12 million COP; Rented: -20 million COP; Automated: -32 million COP                                                                                |
| Fixed, state dependent                | Logistics bonus, no adjustment, or disruption loss from conditions         | $f^S(S_j)$    | Favorable: 8 million COP; Normal: 0 million COP; Adverse: -10 million COP                                                                                       |

Payoff equation.

$$Pay(A_i, S_j) = q(A_i, S_j) [v^A(A_i) + v^S(S_j)] + f^A(A_i) + f^S(S_j).$$

Calculations (million COP).

| Storage strategy           | Favorable                     | Normal                       | Adverse                             |
|----------------------------|-------------------------------|------------------------------|-------------------------------------|
| Contracted storage         | $18(4 + 2) + (-12) + 8 = 104$ | $13(4 + 0) + (-12) + 0 = 40$ | $8(4 + (-2)) + (-12) + (-10) = -6$  |
| Rented warehouse           | $22(5 + 2) + (-20) + 8 = 142$ | $15(5 + 0) + (-20) + 0 = 55$ | $9(5 + (-2)) + (-20) + (-10) = -3$  |
| Automated distribution hub | $26(6 + 2) + (-32) + 8 = 184$ | $17(6 + 0) + (-32) + 0 = 70$ | $10(6 + (-2)) + (-32) + (-10) = -2$ |

**Final payoff table (profit in million COP).**

| Storage strategy           | Favorable | Normal | Adverse |
|----------------------------|-----------|--------|---------|
| Contracted storage         | 104       | 40     | -6      |
| Rented warehouse           | 142       | 55     | -3      |
| Automated distribution hub | 184       | 70     | -2      |